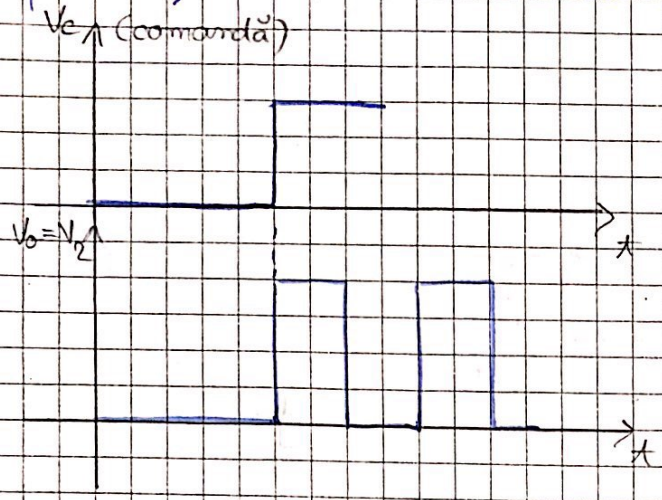
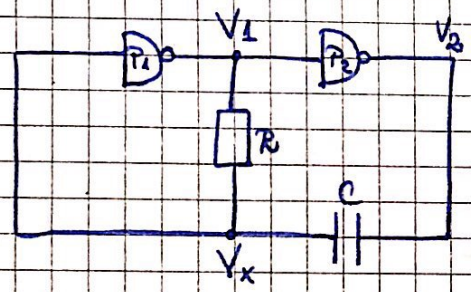
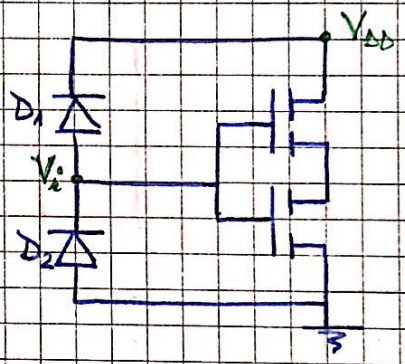
Starea inițială este nedefinită, iar dacă dorim să o știm cu precizie, ne trebuie un semnal de comandă: un reset

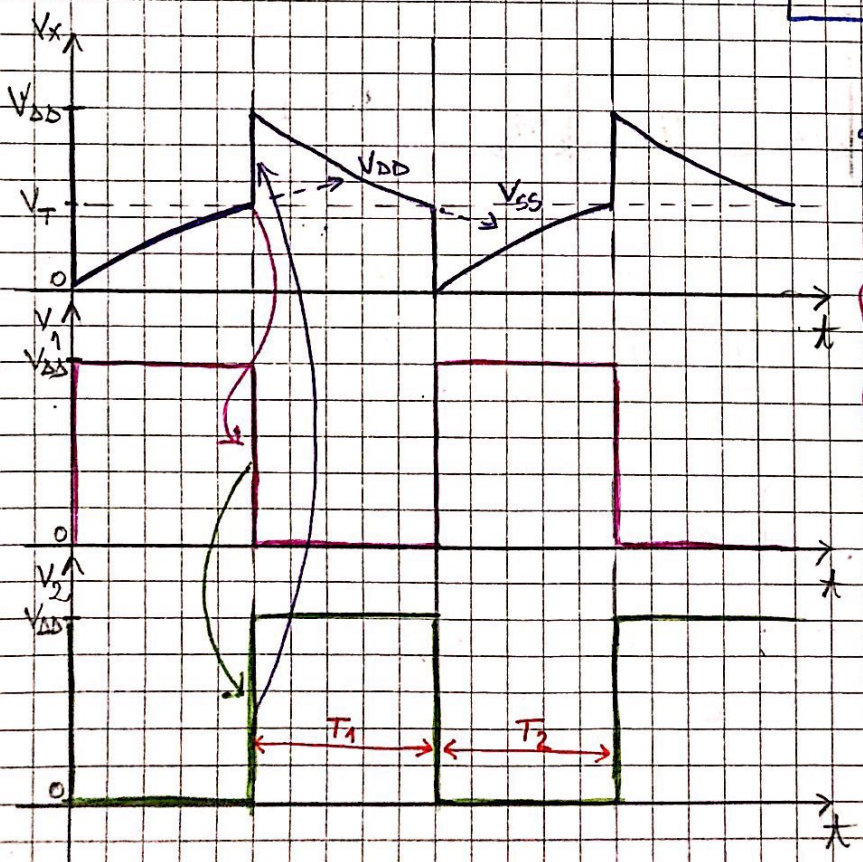


CURS 9 8.05.2019

CBA cu porți CMOS







- Starea inițială este nedefinită

- $T_1 + T_2 = T$

$$T_1 = RC \ln \frac{0 - V_{DD}}{0 - V_T}$$

$$T_2 = RC \ln \frac{V_{DD} - 0}{V_{DD} - V_T}$$

$$\frac{1}{3} V_{DD} \le V_T \le \frac{2}{3} V_{DD}$$

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a) pt. $V_T = \frac{1}{3} V_{DD} \Rightarrow T_1 = RC \ln \frac{0 - V_{DD}}{0 - \frac{1}{3} V_{DD}} = RC \ln 3 \approx 1{,}1\, RC$

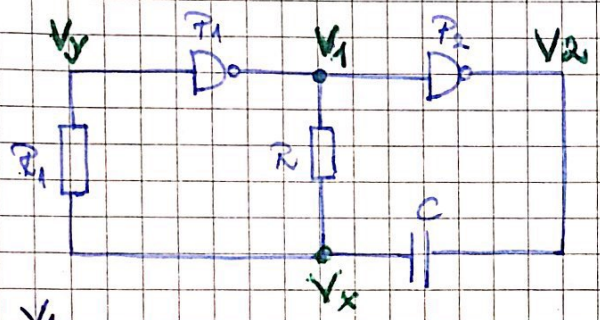
$$T_2 = RC \ln \frac{V_{DD} - 0}{V_{DD} - \frac{1}{3} V_{DD}} = RC \ln \frac{3}{2} \simeq 0{,}4\, RC$$

b) pt. $V_T = \frac{1}{2} V_{DD} \Rightarrow T_1 = T_2 = RC \ln 2 \approx 0{,}7\, RC$

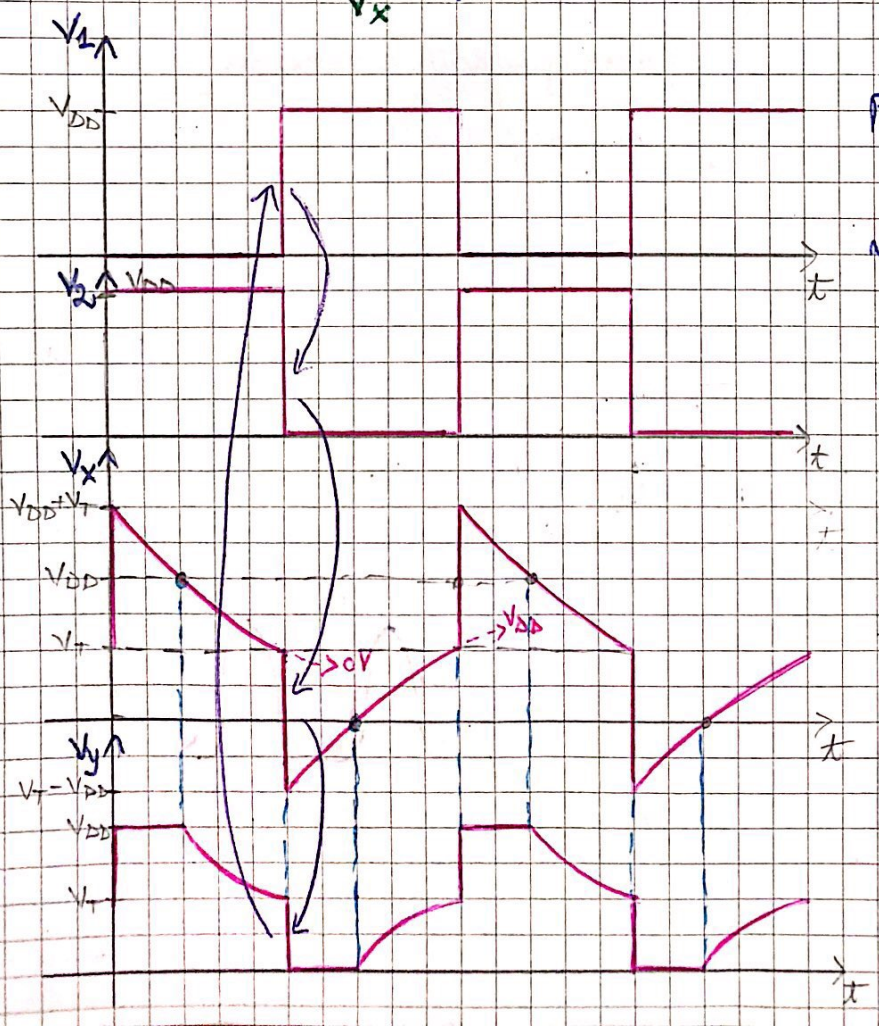
c) pt. $V_T = \frac{2}{3} V_{DD} \Rightarrow T_1 = RC \ln \frac{3}{2} \approx 0{,}4\, RC$

$$T_2 = RC \ln 3 \approx 1{,}1\, RC$$

♥ CBA cu porți CMOS compensat ♥



$R_1 >> R$?



V_x nu e intrare de poartă (avem R_1).

V_y nu scade sub 0 și nu crește peste V_{DD}.

$\therefore T = T_1 + T_2$

fn x: $\begin{cases} T_1 = RC \ln \frac{V_{DD} - (V_T - V_{DD})}{V_{DD} - V_T} \\ T_2 = RC \ln \frac{0 - (V_T + V_{DD})}{0 - V_T} \end{cases}$

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a) pt. $V_T = \frac{1}{3} V_{DD}$

$$T_1 = RC \ln\left(\frac{5}{3}V_{DD} \Big/ \frac{2}{3}V_{DD}\right) = RC \ln \frac{5}{2} \simeq 0{,}91\,RC$$

$$T_2 = RC \ln \frac{\frac{4}{3}V_{DD}}{\frac{1}{3}V_{DD}} = RC \ln 4 \simeq 1{,}3\,RC$$

b) pt. $V_T = \frac{1}{2} V_{DD}$: $T_1 = RC \ln \frac{\frac{3}{2}V_{DD}}{\frac{1}{2}V_{DD}} = RC \ln 3 \simeq 1{,}1\,RC$

$$T_2 = RC \ln \frac{\frac{3}{2}V_{DD}}{\frac{1}{2}V_{DD}} = RC \ln 3 \simeq 1{,}1\,RC$$

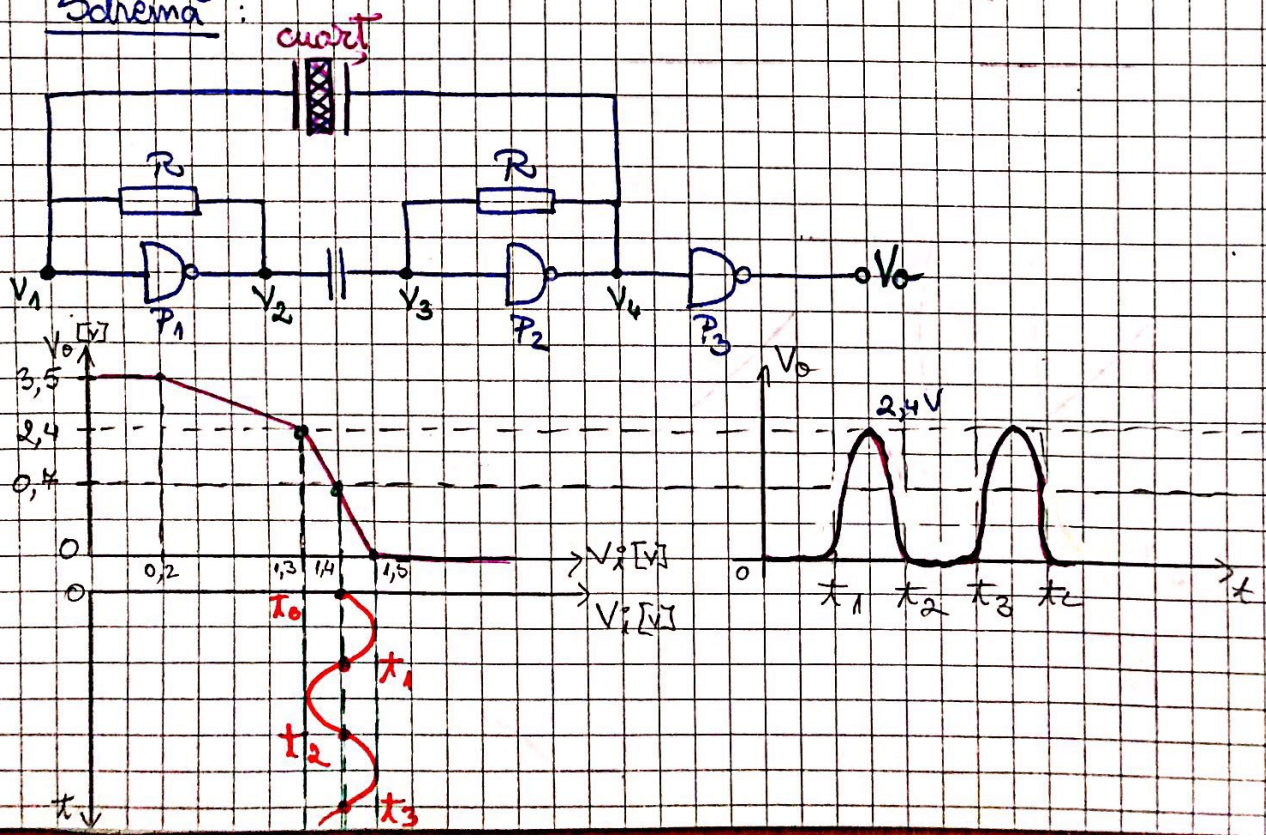
c) pt. $V_T = \frac{2}{3} V_{DD}$: $T_1 = RC \ln 4 \simeq 1{,}3\,RC$

$$T_2 = RC \ln 2{,}5 \simeq 0{,}91\,RC$$

CBA cu cuarț

- cel mai greu circuit
- în fct. de modul și condițiile tăierii, cuarțul are frecv. diferite; face parte din materialele piezo-electrice: generează sarcini electrice sub act. diferitelor forțe (ex: tensiune)
- avantajul cuarțului: frecv. generată e deosebit de constantă => eroare este de p.p.m. (părți per milion)

Schemă:



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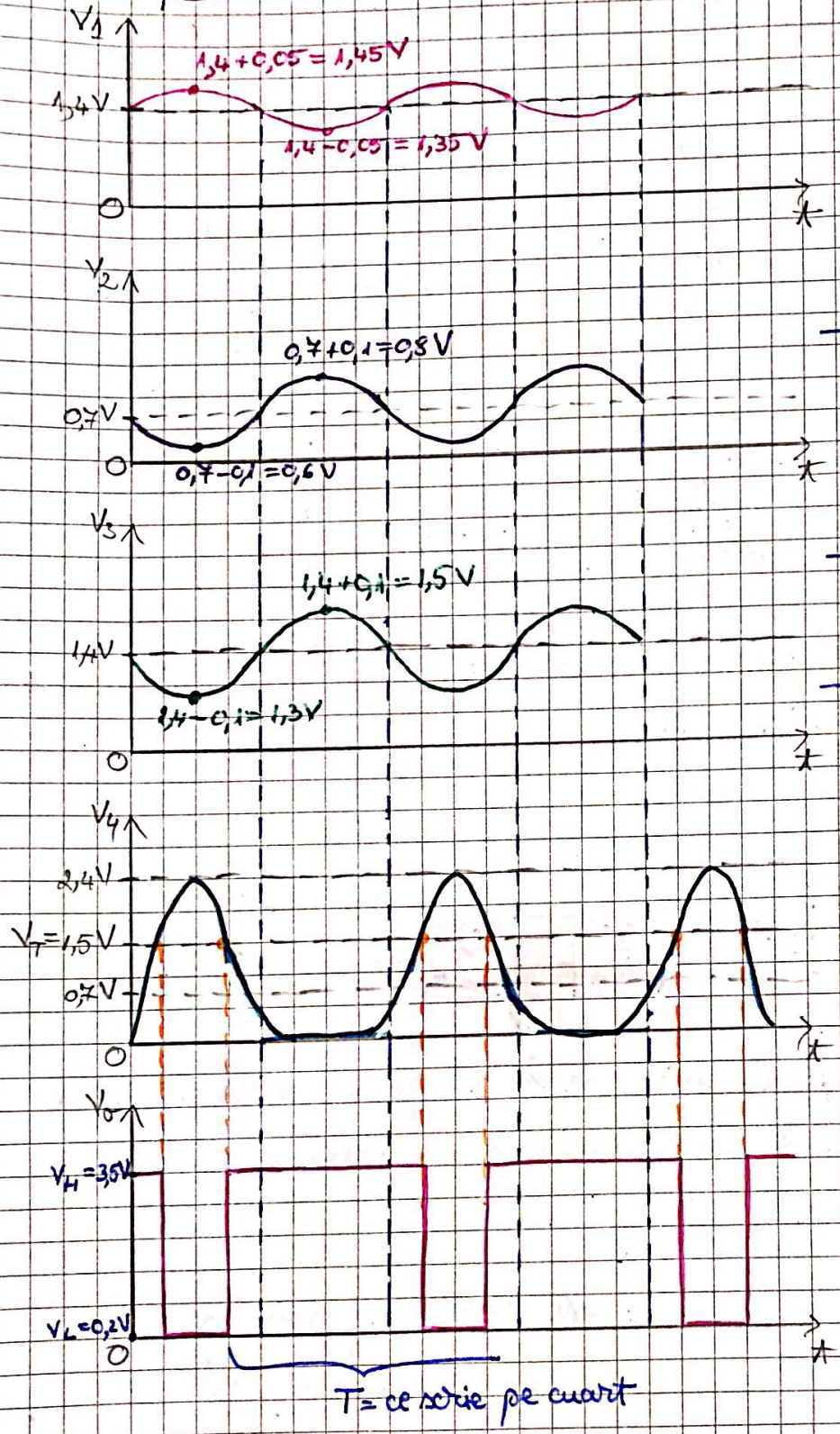
Funcționare:



→ Folosim 2x amplificarea cea mai mică, deci 0,05 => 0,1

→ Capacitatea anulează componenta continuă, dar lasă să treacă sinusoida

→ La V_3, anulăm 0,7 (comp. continuă) și suprapunem sinusoida peste 1,4

→ Cuarțul comandă de fapt tactul

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